



EDUFYI TECH SOLUTIONS

VIRTUAL INTERNSHIPS • INDUSTRY MENTORSHIP

VLSI

MENTORSHIP PROGRAM

RTL • VERIFICATION • PHYSICAL DESIGN • SoC

60

DAYS

8

MODULES

6+

PROJECTS

12+

EDA TOOLS

2026 EDITION / COHORT BROCHURE

WHO WE ARE

At Edufyi Tech Solutions we transform the educational journey of engineering students through cutting-edge virtual internships and structured, industry-mentored training. Every learner ships six real-time projects and one capstone per domain — built on the same tools, flows and sign-off criteria used inside semiconductor design houses.

MISSION

Equip students with virtual internships and real-world silicon projects that convert directly into placements, guided end-to-end by practising industry mentors.

VISION

Bridge academic theory and fab-grade practice, so every graduate enters the semiconductor industry job-ready from day one.

WHY THIS PROGRAM IS DIFFERENT

01

Tool-first learning

Industry EDA flows from week one — not slideware.

02

Mentor pods

Small cohorts reviewed by working RTL, DV and PD engineers.

03

Tape-out mindset

Every lab ends in a signed-off, report-backed deliverable.

04

Placement runway

Resume, GitHub portfolio and mock interview rounds included.

PROGRAM AT A GLANCE

FRONT-END RTL

SystemVerilog, RTL micro-architecture, synthesis QoR, lint & CDC

01

DESIGN VERIFICATION

UVM testbenches, SVA assertions, coverage closure, formal & regression

02

PHYSICAL DESIGN

Floorplan, placement, CTS, routing, STA sign-off, DRC / LVS

03

ANALOG & MIXED-SIGNAL

Op-amps, bandgap, PLL, ADC/DAC, SerDes and layout in Virtuoso

04

SoC & ARCHITECTURE

AMBA AXI/AHB/APB, NoC, RISC-V cores, firmware bring-up

05

DFT & SILICON BRING-UP

Scan insertion, ATPG, MBIST, JTAG, yield and test economics

06

OUTCOME

A GitHub-ready portfolio containing a synthesised RTL block, a UVM verification environment, a placed-and-routed macro with timing sign-off reports, and a documented capstone SoC — plus certification and a letter of recommendation.

01

WEEK

FOUNDATIONS OF VLSI & SEMICONDUCTOR TECHNOLOGY

WHY VLSI, WHY NOW

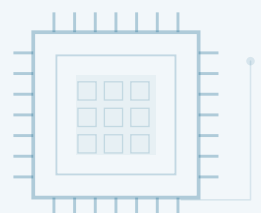
- Moore's Law, Dennard scaling and the More-than-Moore era
- Market map: mobile SoCs, automotive, AI accelerators, aerospace

DEVICE & PROCESS TECHNOLOGY

- MOSFET operation, planar CMOS to FinFET and Gate-All-Around nanosheets
- Wafer fabrication: lithography, EUV, doping, metallisation, packaging

DESIGN ECOSYSTEM

- ASIC vs FPGA vs SoC vs structured ASIC — choosing the right vehicle
- Full VLSI flow: spec, RTL, verification, synthesis, PnR, sign-off, tape-out
- PDKs, standard-cell libraries and the open-source Sky130 flow



02

WEEK

DIGITAL DESIGN & ADVANCED LOGIC ENGINEERING

LOGIC FUNDAMENTALS

- Number systems, Boolean algebra, K-maps, Quine-McCluskey minimisation
- Static CMOS, pass-transistor, transmission-gate and dynamic logic styles

COMBINATIONAL ENGINEERING

- Ripple-carry, carry-look-ahead, carry-select and Wallace-tree multipliers
- Barrel shifters, priority encoders, comparators and ALU datapaths

SEQUENTIAL & TIMING HYGIENE

- Latches vs flip-flops, FSM styles, one-hot vs binary encoding
- Metastability, MTBF, synchronisers and clock-domain-crossing basics
- Memory elements: SRAM, register files, FIFOs and dual-port RAM



03

WEEK

HDL MASTERY: VERILOG, SystemVerilog & VHDL

LANGUAGE FOUNDATIONS

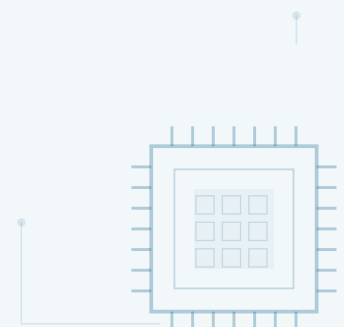
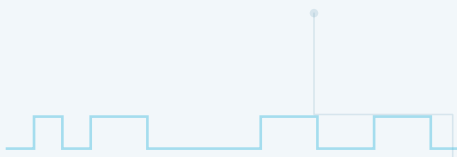
- Verilog / VHDL syntax, data types, operators and hierarchy
- SystemVerilog upgrade: logic, packed arrays, interfaces, packages

DESIGN & MODELLING

- Blocking vs non-blocking, synthesisable subsets, parameterised generics
- Reusable IP with generate blocks, structs, enums and typedefs

SIMULATE & DEBUG

- Self-checking testbenches, tasks, functions, file I/O and randomisation
- Waveform debug in ModelSim / Questa, Xcelium and open-source Verilator
- Lab: parameterised UART, SPI and I2C controllers written from scratch



04

WEEK

RTL DESIGN, SYNTHESIS & QoR OPTIMISATION

MICRO-ARCHITECTURE

- RTL partitioning, pipelining, resource sharing and FSM-datapath split
- Latency vs throughput vs area trade-off studies

LOGIC SYNTHESIS

- Design Compiler / Genus / Yosys: elaborate, map, optimise, report
- SDC constraints: clocks, uncertainty, I/O delay, false and multicycle paths
- Technology libraries, .lib timing arcs, wire-load and Liberty basics

QUALITY OF RESULTS

- Clock gating, operand isolation, retiming and multi-Vt cell selection
- Lint, CDC (Spyglass) and area / power / timing QoR closure loop

05

WEEK

STATIC TIMING ANALYSIS & FUNCTIONAL VERIFICATION

TIMING THEORY

- Setup, hold, recovery, removal, clock skew, jitter and propagation delay
- Path types, slack computation, OCV / AOCV / POCV and derates

STA IN PRACTICE

- PrimeTime / Tempus flow, MMMC corners, SDF and timing ECOs
- Reading and fixing timing reports; hold-fix strategies

VERIFICATION ENGINEERING

- Coverage-driven verification, constrained random stimulus, scoreboards
- UVM architecture: agent, driver, monitor, sequencer, sequences
- SystemVerilog Assertions, functional coverage and formal property checking

06

WEEK

ANALOG, MIXED-SIGNAL & HIGH-SPEED CIRCUITS

ANALOG BUILDING BLOCKS

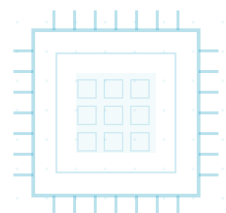
- Current mirrors, differential pairs, two-stage and folded-cascode op-amps
- Bandgap references, LDOs, comparators, active filters and oscillators

DATA CONVERTERS

- Sampling theory, quantisation noise, SNR / SFDR / ENOB metrics
- SAR, flash, pipeline and sigma-delta ADCs; R-2R and current-steering DACs

HIGH-SPEED & LAYOUT

- PLL / DLL architecture, jitter budgeting, SerDes and equalisation basics
- Cadence Virtuoso schematic entry, Spectre simulation, layout, DRC / LVS
- Noise, matching, ESD and mixed-signal floorplanning discipline



07

WEEK

ADVANCED SoC, LOW-POWER, DFT & PHYSICAL DESIGN

LOW-POWER DESIGN

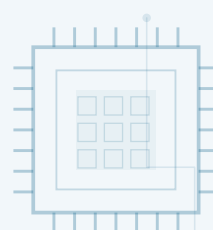
- Dynamic, short-circuit and leakage power; power intent with UPF / CPF
- Clock gating, power gating, multi-Vdd, DVFS, isolation and retention cells

SOC ARCHITECTURE

- AMBA AHB / APB / AXI4 / ACE protocols, interconnect and Network-on-Chip
- Processor subsystems, cache coherency, DMA, memory controllers, boot flow

DFT & PHYSICAL IMPLEMENTATION

- Scan insertion, ATPG, fault models, MBIST, JTAG / boundary scan
- Floorplanning, power planning, placement, CTS, routing in Innovus / ICC2
- Signal integrity, IR-drop, EM, DRC / LVS / antenna sign-off



08

WEEK

CAPSTONE, EMERGING SILICON & CAREER LAUNCH

CAPSTONE BUILD

- Scope, plan and execute an RTL-to-GDSII or full UVM verification project
- Design reviews, regression reports, documentation and defence presentation

FRONTIER TECHNOLOGIES

- 3D ICs, chiplets and UCIe; advanced packaging (CoWoS, InFO)
- AI/ML accelerators, systolic arrays, in-memory and neuromorphic compute
- RISC-V ecosystem, open-source EDA, and AI-assisted chip design

CAREER LAUNCH

- Domain-wise resume, GitHub portfolio and LinkedIn positioning
- Mock interviews: RTL puzzles, STA questions, protocol and system design rounds

ADVANCED ELECTIVES

Choose two specialisation add-ons in the final fortnight to deepen your niche.

01

RISC-V SoC BRING-UP

Integrate a RV32I core, memory map, peripherals and run bare-metal firmware.

02

UVM ADVANCED

Layered sequences, RAL model, callbacks, virtual sequencers, regression triage.

03

RTL-to-GDSII SPRINT

Full OpenLane / Innovus implementation of a block with sign-off reports.

04

AI ACCELERATOR DESIGN

Systolic MAC array, dataflow mapping, quantisation and memory bandwidth.

05

HIGH-SPEED INTERFACES

DDR, PCIe and SerDes fundamentals with equalisation and eye diagrams.

06

FPGA PROTOTYPING

Vivado flow, timing closure on FPGA, ILA debug and hardware validation.

INDUSTRY TOOL STACK

Questa / ModelSim

Xcelium

Verilator

Design Compiler

Genus

Yosys

PrimeTime

Innovus

ICC2

Virtuoso

Spectre

Vivado

Spyglass

OpenLane / Sky130

PROJECT PORTFOLIO

Six real-time builds plus one defended capstone.

P1

16-bit Pipelined RISC Datapath

RTL micro-architecture, hazard handling, synthesis QoR report.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

P2

AXI4-Lite Slave with UVM Testbench

Protocol compliance, constrained random, coverage closure.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

P3

Low-Power Clock-Gated FIR Filter

UPF intent, power estimation, multi-Vt optimisation.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

P4

SAR ADC in Cadence Virtuoso

Schematic, Spectre sims, layout, DRC/LVS clean.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

P5

RTL-to-GDSII of a Crypto Block

Floorplan, CTS, routing, STA and physical sign-off.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

P6

DFT Scan + MBIST Insertion

Scan chains, ATPG coverage report, test-time analysis.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

CAP

Capstone: RISC-V SoC or Full DV Environment

End-to-end build, documentation, design review and defence.

[deliverable](#) · [source](#) + [reports](#) + [walkthrough video](#)

HOW IT WORKS

- 1 Schedule a demo call**
Explore the curriculum, mentors and outcomes with our team.
- 2 Choose your program**
Pick the track and plan that matches your goals and budget.
- 3 Enroll & onboard**
Get tool access, mentor pod allocation and your project charter.
- 4 Engage for 2-3 months**
Weekly live mentorship, labs, reviews and regression checkpoints.
- 5 Graduate with proof**
Portfolio, certificates and interview readiness — ready to be hired.

WHAT YOU RECEIVE



Internship Certificate



Training Certificate



Course Certificate



**Letter of
Recommendation**

RECOGNITION

Approved and credentialed by

Ministry of Corporate Affairs

MSME

ISO 9001:2015

AICTE

Wipro — Credential Partner

Startup India

MENTORS FROM LEADING SEMICONDUCTOR COMPANIES

INTEL

NVIDIA

QUALCOMM

CADENCE

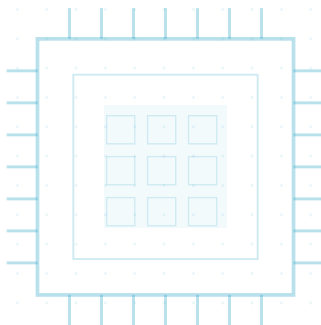
SYNOPSYS

AMD

TEXAS INSTRUMENTS

BOSCH

MICRON



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Azure Fundamentals



IBM
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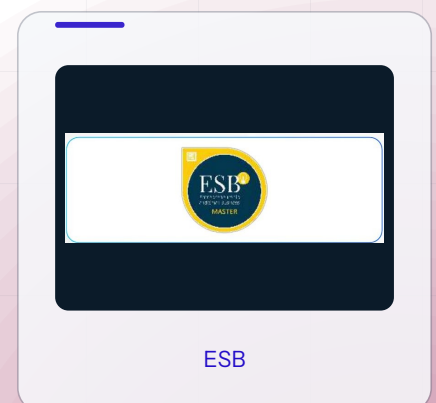
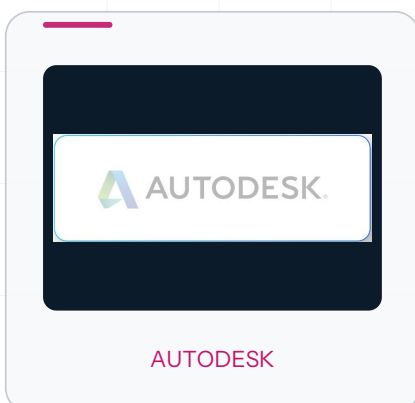
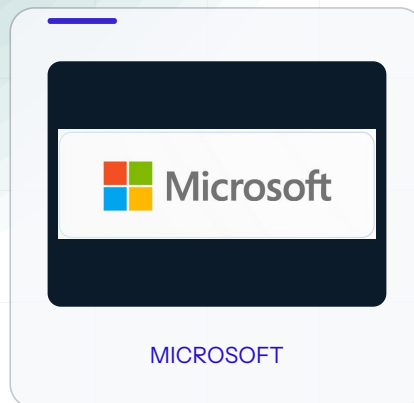
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